



NITI Aayog

SECTOR AT A GLANCE 04

Bio Energy in India

Energy in India

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The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Bioenergy, a form of renewable energy, is derived from biomass and feedstock, including agricultural residues, forestry residues, animal waste, and organic municipal solid waste. The biomass and feedstock are harnessed through processes like combustion, anaerobic digestion, and biochemical conversion to generate bioenergy in the form of (1) biopower or biomass power (for electricity); (2) biofuels such as ethanol, sustainable aviation fuel, and biodiesel (usually for transportation); (3) biogas and briquettes (for cooking); and (4) waste to energy. Bioenergy (biomass power and co-generation) accounts for 2.5% of the total installed electricity capacity and 6% of the renewable energy mix as of 2023.¹ Approximately 10.4% of the country's total primary energy comes from biomass, and over 70% of the population relies on it for their energy requirements.²

Scaling efforts in bioenergy adoption can accelerate India's transition to renewable energy, replacing traditional biomass for cooking with cleaner cooking fuels, increasing the use of biofuels in transportation, meeting energy needs in remote and rural areas, facilitating organic waste management, and contributing to the green jobs market.

Harnessing the full potential of the bioenergy sector requires overcoming challenges such as limited feedstock availability, variable feedstock quality, expensive key bioenergy raw materials, constrained ethanol distribution, and high tariffs for biomass power generation.

Given these opportunities and challenges, the installation of large biogas, bio-CNG power plants, and increasing production of ethanol are the major policy focus areas. Several ongoing and recent central government initiatives, including but not limited to the National Bioenergy Programme and National Policy on Biofuels focus on providing financial and technical assistance to set up these power plants. These efforts along with several state- and local-level policies and initiatives, seek to enhance India's energy security while providing social and environmental benefits.

Investing in critical public infrastructure across the bioenergy supply chain, promoting more and diverse use cases, catalyzing innovative financing, and ensuring adequate feedstock supply will remain critical to achieve India's bio energy goals.

This note presents key trends related to different components of bio energy in India and actions for the way forward.

Figure 1: Key bioenergy statistics

2.5%
Bio power's share in India's total installed electricity generation capacity

10 GW
Installed biopower capacity in India as of 2023

42 GW
Bio power potential from surplus biomass and bagasse-based cogeneration in sugas mills

20%
Ethanol blending in petrol target set for 2025

Source: Ministry of Power, Power sector at a glance ALL India, accessed on December 2023; Ministry of New and Renewable Energy, Bio Energy, 2022; Ministry of Petroleum and Natural Gas, India has achieved the target of 10 percent ethanol blending, 5 months ahead of schedule, 2022

¹ Ministry of Power, Power Sector at a Glance ALL INDIA, 2023

² Ministry of New and Renewable Energy, Bio Energy, 2022; NITI Aayog, India Energy Security Scenarios 2047 (as under Least Effort Scenario – All Level 1), Accessed June 2024

Key trends

This section dives deeper into India's bio energy landscape, outlining statistics and trends on:

1. Resourcing, planning, and data management
2. Biomass sourcing and pre-treatment
3. Generation, integration, and production
4. Distribution and marketing
5. Bioenergy utilisation
6. Workforce development
7. Monitoring and evaluation

Resourcing, planning, and data management

India sees massive growth in bio energy and has set ambitious 2030 targets.

India's biopower potential is 42 GW from surplus biomass and bagasse-based cogeneration in sugar mills:

A study by the Ministry of New and Renewable Energy (MNRE) estimates that about 230 million metric tonnes of surplus biomass are available in India every year. This surplus biomass could be used to generate about 28 GW of biopower. Additionally, there is a presence of 550 sugar mills across the country, which can be technically adapted to generate an additional 14 GW of power through bagasse-based cogeneration.³

India's compressed biogas (CBG) potential is 62 million metric tonnes (MMT), and targets have been set to achieve 15 MMT of CBG by 2024-25:

As per the MNRE, India can produce 62 MMT of CBG if the full potential of compressed CBG is exploited which is sufficient to fulfil the whole natural gas demand of the country. India has set the target to set up 5,000 commercial plants to produce 15 MMT of CBG by 2024-25 under the Sustainable Alternative Towards Affordable Transportation (SATAT) scheme.⁴

India has set ambitious targets for biofuels to be achieved by 2025-26:

Under the National Biofuels Policy 2018, targets for 20% ethanol blending in gasoline and 5% biodiesel blending in diesel were set, and India is well on track to achieve the set targets given 12% blending of ethanol has already been achieved by 2022.⁵ Under India's G20 presidency Global Biofuel Alliance was one of the priorities through which it aimed to strengthen markets, facilitate global biofuel trade, and development of concrete policy lesson-sharing mechanisms.⁶

Biomass sourcing and pre-treatment

Biomass and feedstock such as agricultural residues, oil seeds, organic waste, municipal solid and liquid waste, etc., are key raw materials to produce bioenergy (including biofuels, biogas, and biopower). Despite the abundant availability of raw materials to produce bioenergy, several challenges and uncertainties exist relating to biomass supply, fuel supply and quality, and market entry.

India has availability of a diverse variety of biomass and feedstock to produce bioenergy:

A study by the Ministry of New and Renewable Energy (MNRE) estimated that there are about 750 million metric tonnes of biomass available in India each year, of which about 230 million metric tonnes are surplus including products such as bagasse, rice husk, straw, cotton stalk, coconut shells, soya husk, de-oiled cakes, coffee waste, jute wastes. 550 sugar mills across the country are available for bagasse-based

³ Ministry of New and Renewable Energy, Bio Energy, 2022

⁴ Ministry of Petroleum and Natural Gas, Government has set target to increase share of gas in energy mix up to 15 per cent by 2030 : Union Minister of Petroleum & Natural Gas and Housing & Urban Affairs Shri Hardeep Singh Puri, 2023; Centre for Science and Environment, COMPRESSED BIOGAS (CBG) : Potential, Technology, Policy, Operations and Economics, 2023

⁵ Ministry of Consumer Affairs, Food & Public Distribution, Ethanol Production Capacity in the country is 1380 crore litres of which 875 crore litres is molasses based and 505 crore litres is grain based, 2023

⁶ Ministry of Petroleum & Natural Gas, Global Biofuel Alliance: One of the priorities under India's G20 Presidency, 2023

cogeneration of biopower.⁷ In addition, about 60 million tonnes of organic waste from urban municipal areas were available as of 2021.⁸ Further, 178 million tonnes of dry biomass from eleven major crops are available for producing cellulosic ethanol (2nd Generation – 2G), 300 kilotonnes of used cooking oil, and 1.22 million tonnes of Tree Borne Oil (TBO) seed species (that can be used to produce biodiesel or bio-jet fuels).⁹ Under the Biomass programme the Government of India support setting up of Biomass Briquette/Pellet manufacturing plants and to support Biomass (nonbagasse) based cogeneration projects in Industries in the country. The broader objectives of the scheme are to reduce stubble burning by utilizing surplus agricultural residue.¹⁰

The government has expanded its view on the types of feedstocks that can be utilized as raw materials to produce ethanol biofuel:

The National Biofuels Policy 2018 added sugarcane molasses, sugarcane juice, damaged food grains, used cooking oil, sugar beet, starch-containing materials like corn, cassava that are unfit for human consumption – as new potential sources to produce ethanol. This aims to build momentum for achieving India's biofuels targets set for 2025-26.¹¹

Generation, integration, and production

India has made progress in boosting power generation from bioenergy and increasing ethanol blending capacity. However, challenges remain, including the high cost of biomass storage, high upfront costs, lack of financing options, issues in obtaining long-term contracts with DISCOMs, and the lack of affordable and compact biomass systems for decentralized energy generation in rural areas.

India's biopower generation capacity in 2023 has increased ~100% from 2015-16:

India's biopower generation capacity as of May 2023 is 10.2 GW, which has grown ~2X from 4.83 GW in 2015-16. India set a target to achieve 10 GW of biopower by 2022 in its climate commitment in 2015.¹²

India's biofuel production (including biodiesel) has increased substantially in the last two decades:

Bioethanol production has increased ~20X from 15.4 crore litres in 2012-13 to 304 crore litres in 2020-21.¹³ Additionally, the production of biodiesels increased ~2X from 10 crore litres in 2010-11 to 18.5 crore litres in 2022-23.¹⁴

The central government provides targeted financial assistance to incentivize bioenergy projects:

In November 2022, the Ministry of New and Renewable Energy (MNRE) unveiled the National Bioenergy Programme, allocating a budget of INR 1715 crores for FY 2021-22 to 2025-26. Phase 1 of the project has a specific budget of INR 858 crores to provide central financial assistance (CFA) through three programs – (I) Waste to Energy Programme (INR 600 crores), Biomass Programme (INR 158 crores) and Biogas Programme (INR 100 crores) to companies involved in biogas, bio-CNG, electricity production from waste sources, biomass briquette/pellet manufacturing, and non-bagasse biopower cogeneration.¹⁵ Additionally, under the Pradhan Mantri JI-VAN Yojana, a total budget of ~INR 1970 crores has been set up for the period 2018-19 to 2023-24 to provide financial support for setting up 2nd Generation (2G) ethanol plants in the country using lignocellulosic biomass and renewable feedstock.¹⁶ In the 2023 Union Budget, the central government also announced a budget of INR 1,000 crores to set up 200 compressed biogas plants and 300 community or cluster-based plants under the GOBARDhan scheme.¹⁷

7 Ministry of New and Renewable Energy, Bio Energy, 2022

8 International Energy Agency, India Energy Outlook 2021, 2021

9 International Energy Agency, Commercializing Conventional and Advanced Transport Biofuels from Biomass and Other Renewable Feedstocks, 2020

10 Ministry of New and Renewable Energy, Implementation of Biomass Programme, 2022

11 International Energy Agency, Implementation of Bioenergy in India, 2021

12 Ministry of Power, Power Sector at a Glance, 2023; Ministry of New and Renewable Energy, Renewable Energy in India, 2022; PRS, Action plan for achievement of 175 GW renewable energy target, 2021

13 ORF, The Implications of India's Revised Roadmap for Biofuels: A Lifecycle Perspective, 2022

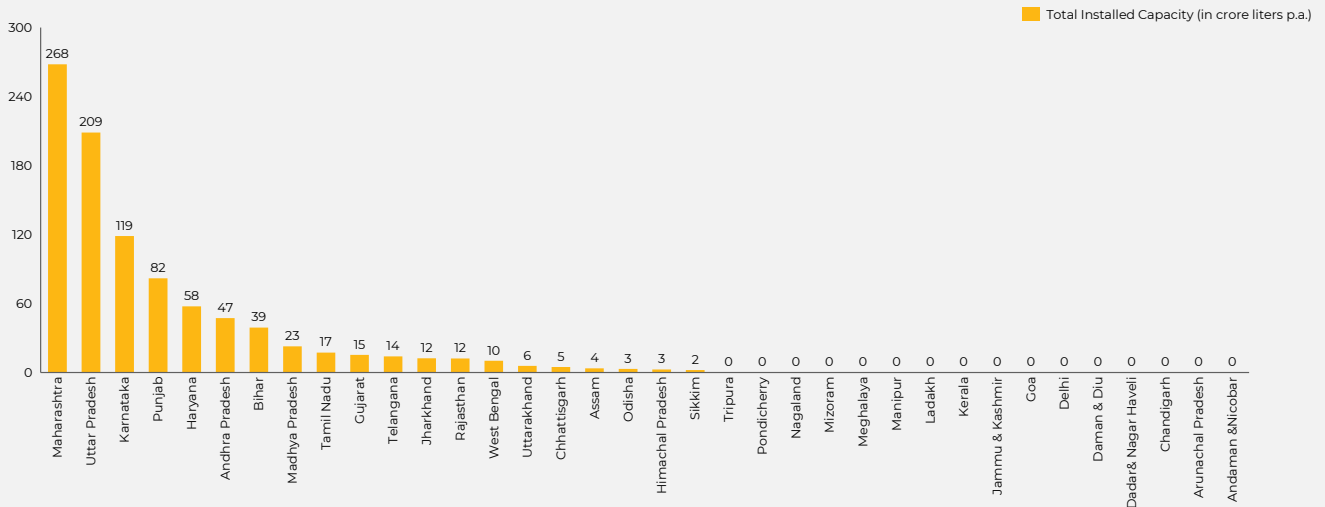
14 Ministry of Petroleum and Natural Gas, Biodiesel, 2022

15 Ministry of New and Renewable Energy, Ministry of New and Renewable Energy initiates National Bio Energy Programme to utilize surplus biomass for power generation, 2022; MNRE, MNRE notifies National Bio Energy Programme, 2022

16 Ministry of Petroleum and Natural Gas, The Government had notified the Pradhan Mantri JI-VAN Yojana for providing financial support to integrated bio-ethanol projects, 2022

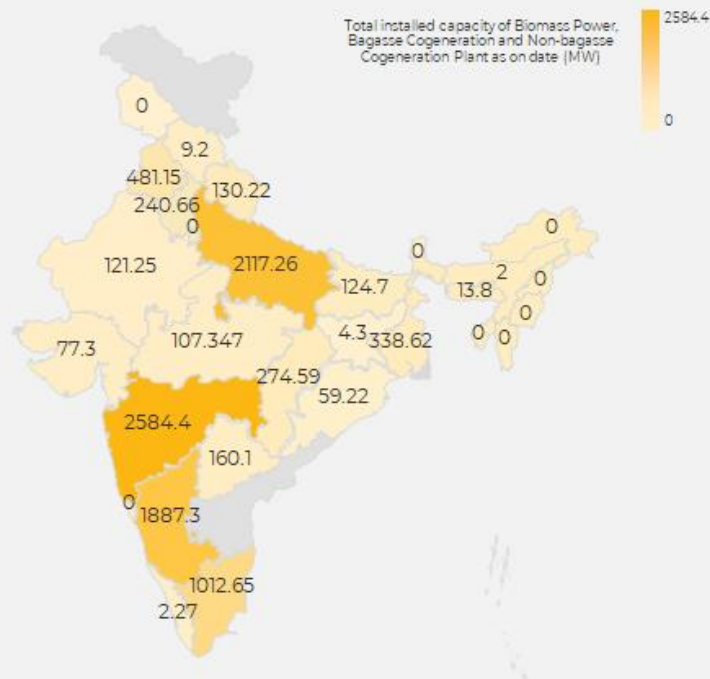
17 PwC India, Fuelling India's future with bioenergy, 2023

Figure 2: Installed ethanol production capacities by state and UT (2022)



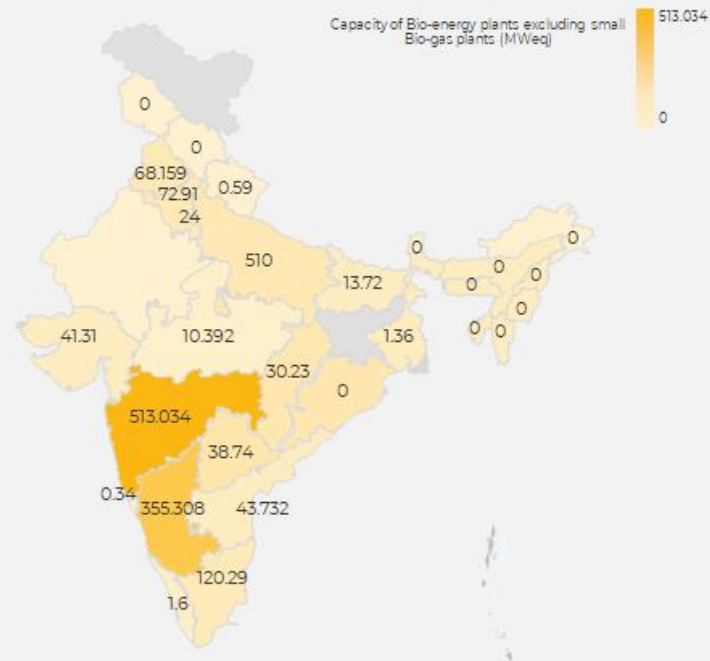
Source: Ministry of New and Renewable Energy, current ethanol production capacity for ethanol blending with petrol and other uses, 2022

Figure 3: Biomass power and bagasse/non-bagasse cogeneration projects installed by state and UT (2023)



Source: Ministry of New and Renewable Energy, 10,232 MW of installed capacity of Biomass Power, Bagasse Cogeneration and Non-bagasse Cogeneration Plant in India with Maharashtra and UP accounting for nearly 45% of installed capacity: Union Power & NRE Minister Shri R. K. Singh, 2023

Figure 4: Bioenergy plants established in past five years (from FY 2017-18 to FY 2021-22) by state and UT



Source: Ministry of New and Renewable Energy, 10,232 MW of installed capacity of Biomass Power, Bagasse Cogeneration and Non-bagasse Cogeneration Plant in India with Maharashtra and UP accounting for nearly 45% of installed capacity: Union Power & NRE Minister Shri R. K. Singh, 2023

Distribution and marketing

The National Policy on Biofuels, 2018 has laid down guidelines around biofuel distribution, marketing, and pricing. The Ministry of Petroleum and Natural Gas has established the pricing mechanism for Compressed Biogas (CBG). Space constraints, and lack of policy implementation, are major challenges in the distribution and marketing of bioenergy.

OMCs handle biofuel storage, distribution, and marketing:

As per the National Policy on Biofuels (2018), Oil Marketing Companies (OMCs) are primarily responsible for maintaining and improving the storage, distribution, and marketing infrastructure to meet the requirements of biofuels.¹⁸

The central government and OMCs are responsible for biofuel pricing:

Currently, the central government determines the price of first-generation molasses-based ethanol for the Ethanol Blended Petrol Programme through a dedicated committee, while OMCs determine the price of biodiesel for blending in diesel. The government incentivizes first-generation biofuels through administered or market-based prices, considering factors like market conditions, domestic availability, and import substitution. National Policy on Biofuels provides for differential pricing of advanced biofuels.¹⁹

The government is actively working to enhance the adoption of biogas in rural areas:

The New National Biogas and Organic Manure Programmes (NNBOMP) promotes the use of biogas produced from cattle manure and organic wastes in rural areas. It has successfully established small-scale biogas plants with capacities ranging from 1 to 25 cubic meters (m³) per day. The program benefits beneficiaries by converting cattle dung and other organic wastes into clean gaseous fuel for cooking and lighting, and additionally, it produces nutrient-enriched organic fertilizer as a by-product of biogas generation.²⁰

¹⁸ Ministry of Petroleum and Natural Gas, National Policy on Biofuels, 2018

¹⁹ Ministry of Petroleum and Natural Gas, National Policy on Biofuels, 2018

²⁰ Ministry of New and Renewable Energy, Bio-energy schemes, accessed July 2023

Bioenergy utilization

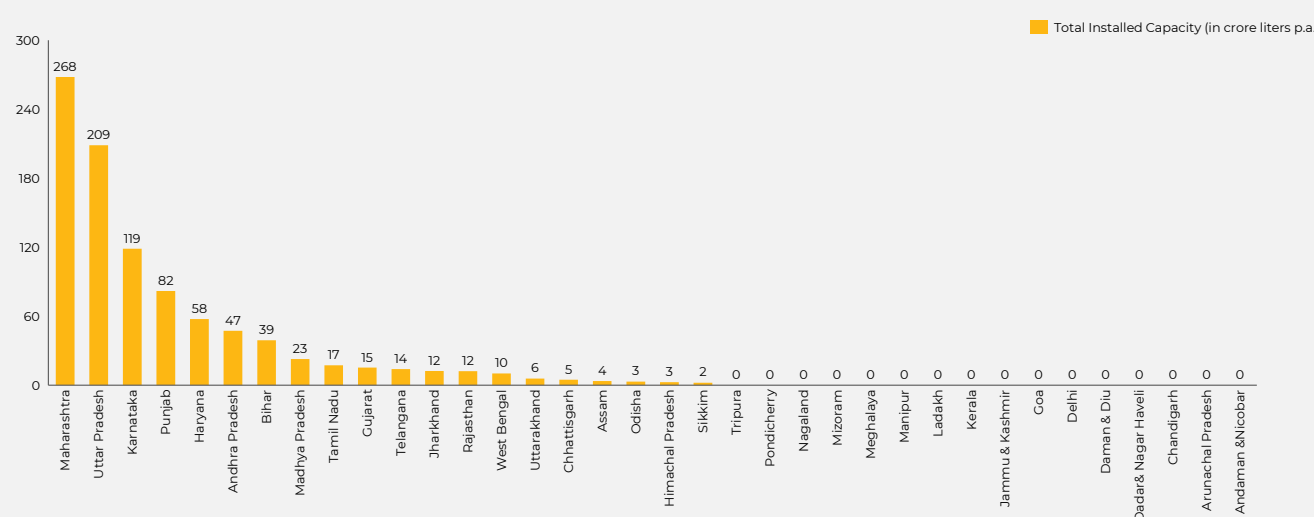
Bioenergy has a wide range of applications across the transport, electricity, and heat/fuel sectors but high tariffs for electricity from biomass power plants and limited offtake of bio-manure and fertilizers, which are key by-products of biogas generation plants, hinder its full potential.

The adoption of biofuels has been low in India:

Biofuels, primarily bioethanol, account for approximately 1% of the total energy consumed in the transportation sector in India. The utilization of biofuels in the transportation sector has started to increase owing to government efforts like the Sustainable Alternative Towards Affordable Transportation (SATAT) scheme to support the development of alternate clean fuels like CBG in transportation, industrial, and commercial uses.²¹ The expected ethanol demand in transportation to meet the 2025 target of 20% ethanol blending is 1016 crore litres.²²

Government initiatives are instrumental in promoting bioenergy utilization for electricity: The contribution of biomass to total installed capacity is about 2.5% in 2023-24, as of November 2023.²³ The Amended National Tariff Policy-2016 mandates Distribution Licensee(s) to purchase 100% of the power generated by Waste-to-Energy plants in the state. This can benefit biogas plants as well.²⁴

Figure 5: Installed ethanol production capacities by state and UT (2022)



Source: Ministry of New and Renewable Energy, The current ethanol production capacity for ethanol blending with petrol and other uses is about 947 crore litres (which includes 619 crore litres of molasses based production capacity and 328 crore litres of grain based production capacity), 2022

Bioenergy is a major source of heat and fuel, particularly in the residential sector:

Bioenergy accounts for about 44% of the total energy consumed for heat and fuel purposes. This encompasses various sectors such as industry, residential, commercial and public services, agriculture, and forestry. In the industrial sector, where fossil fuels take precedence, bioenergy represents 20% of fuel consumption.²⁵

²¹ IEA Bioenergy, Implementation of bioenergy in India, 2021

²² NITI Aayog, Roadmap for ethanol blending in India 2020-25, 2021

²³ Central Electricity Authority, Installed Capacity Report, 2023

²⁴ IEA Bioenergy, Implementation of bioenergy in India, 2021

²⁵ IEA Bioenergy, Implementation of bioenergy in India, 2021

Research and development

The Research and Development in the renewable energy sector of India are focused less on bioenergy; however, the trend is changing in recent years through initiatives such as Pradhan Mantri JI-VAN Yojana.

The central government's R&D efforts are concentrated on solar and hydrogen energy:

Only approximately 5% of the total renewable R&D expenditure, i.e., INR 22 crores from 2012 to 2016, was spent on bioenergy, which is extremely low in comparison to ~INR 42 crores in hydrogen energy and ~INR 180 crores in solar energy.²⁶

Current bioenergy R&D efforts are focused on the development on 2nd and next generation of biofuels:

After the National Biofuels Policy release in 2018, the Indian government has been promoting research on advanced biofuels such as 2G ethanol, algal biofuel, bio-hydrogen, bio-butanol, and drop-in fuels. The focus is on developing new raw materials, refining processing and conversion technologies, and supporting dedicated bioenergy centres, capacity building, and infrastructure. The central government has also launched a USD 300 million "Pradhan Mantri JI-VAN Yojana" to support commercial and demonstration projects in the field of advanced biofuels.²⁷

The government of India has undertaken research efforts in recent years for the pre-treatment of biomass:

Through the National Institute of Bioenergy, MNRE has funded research to develop techniques for the pre-treatment of biomass, such as agricultural residues, which can improve the efficiency of biomass conversion into bioenergy and increase the overall energy yield.²⁸

Workforce development

The bioenergy sector in India is poised to become the largest employer in India in the renewable energy space; however, challenges related to equal representation of women in the labor force persist, which needs to be solved.

Biomass energy is expected to create 2 million jobs by 2050:

The bioenergy sector in India will be the largest employment generator in the renewable energy space after solar energy.²⁹

Bioenergy sector generates twice the number of white-collar jobs vis-à-vis wind and solar energy:

As per estimates, the bioenergy sector creates 12-18 direct white-collar jobs per Megawatts of energy generation in comparison to 5-10 and 7-8 jobs created in solar and wind energy sectors respectively. This is driven by the highly labor-intensive nature of the bioenergy sector, especially due to the requirement of an additional workforce to manage biomass fuel sourcing and processing.³⁰

Capacity building initiatives for bioenergy:

The Skill Council for Green Jobs provides training and certifications in key bioenergy sectors such as biomass power and co-generation, waste to energy, as well as biofuels and biogas.³¹

Monitoring and evaluation

²⁶ Ministry of New and Renewable Energy, Funds for R&D of New and Renewable Energy Technology, 2016

²⁷ Ministry of Petroleum and Natural Gas, National Policy on Biofuels, 2018; International Energy Agency, Implementation of bioenergy in India, 2021

²⁸ Ministry of New and Renewable Energy, Annual Report 2022-23, 2023; International Energy Agency, Biomass pre-treatment for bioenergy, 2019

²⁹ CEEW, Future skills and job creation with renewable energy in India: Assessing the cobenefits of decarbonising the power sector, 2019

³⁰ PwC, Fuelling India's future with bioenergy, 2023

³¹ Skill Council for Green Jobs, Key Sectors under SCGJ, accessed August 2023

The Sardar Swaran Singh National Institute of Bioenergy (NIBE) under the Ministry of New and Renewable Energy (MNRE) is the principal monitoring and evaluation body for bioenergy energy projects in India, alongside support from State Nodal Agencies.

The GoI employs a dual set-up of inspection agencies for performance monitoring of its bioenergy projects:

At the national level, the concerned State Nodal Agencies for Renewable Energy (SNAs) and the Sardar Swaran Singh National Institute of Bioenergy (SSS-NIBE) act as the performance inspection agencies for the Scheme to Support Manufacturing of Briquettes and Pellets and Promotion of Biomass (non-bagasse) based cogeneration.³²

International collaboration

India is actively engaging in global partnerships and initiatives to promote sustainable biofuels advance clean energy solutions, deepen India's focus on biofuels in the energy mix, and accelerate R&D efforts.

India is setting up a Global Biofuels Alliance to promote the international use of sustainable biofuels:

India, under its G20 presidency, has partnered with Brazil, the United States, and other key stakeholder countries in the biofuel industry to establish a Global Biofuels Alliance, which aims to promote collaboration and enhance the use of sustainable biofuels, particularly in transportation. It will focus on strengthening markets, facilitating global biofuels trade, sharing policy lessons, and providing technical support for biofuels programs worldwide while emphasizing successful practices. The alliance will work alongside existing initiatives such as the Clean Energy Ministerial Biofuture Platform, the Mission Innovation Bioenergy initiatives, and the Global Bioenergy Partnership (GBEP).³³

India has partnered with the International Energy Agency to expand its biofuel production and learn from the R&D efforts of other countries:

India joined International Energy Agency's Technology Collaboration Programme (TCP) on Bioenergy as its 25th member in 2019. Through this partnership, India aims to engage in international collaboration to drive research, develop innovative bioenergy technologies, and analyze policies to overcome various barriers, such as environmental concerns, institutional challenges, technological limitations, etc., that impede the widespread implementation of bioenergy technologies.³⁴

India has also partnered with several countries to drive innovative efforts in the bioenergy space:

In collaboration with the United States, the GoI has set up an "Innovation for Sustainable Aviation Fuel" (ISAF) initiative within Mission Innovation 2.0 to foster global partnerships to develop sustainable and cost-effective strategies for sustainable aviation fuel through global research, private sector investment, and information sharing. At COP26, India also launched the "Integrated Biorefineries" mission with the Netherlands to develop clean energy solutions. Integrated biorefineries are facilities that transform biomass into value-added products while maximizing energy recovery.³⁵

³² Ministry of New and Renewable Energy, Bio-Energy Schemes, accessed July 2023

³³ PIB, Global Biofuel Alliance: One of the priorities under India's G20 Presidency, 2023

³⁴ PIB, Cabinet approves joining of IEA Bioenergy TCP by Ministry of Petroleum and Natural Gas as a Member, 2019

³⁵ International Energy Agency, Implementation of bioenergy in India, 2021; Mission Innovation, Innovation for Sustainable Aviation Fuels; ORF, The Perquisite of Having a Robust Biofuel Industry in India. 2021

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by adopting effective bio energy practices. This primarily involve the potential of biofuels in reducing vehicular emissions, the transition of a large population to cleaner cooking fuels, and the employment-generation potential of the bioenergy industry.



Accelerate India's RE transition: 230 million metric tonnes (MMT) of surplus biomass are available in India annually, which could generate about 28 GW of biopower. Additionally, there is a presence of 550 sugar mills across the country, which can be technically adapted to generate an additional 14 GW of power through bagasse-based cogeneration.³⁶



Provision of clean cooking fuels and electricity in rural areas: In rural areas, there exists an opportunity to replace traditional fossil fuels partially or completely with bioenergy solutions. Specifically, clean cooking fuels can be achieved through the application of biogas, while electricity needs can be met by employing biomass gasifiers. Moreover, the substantial potential for setting up biogas plants is evident when considering the sizeable livestock population of 512 million.³⁷



Increased uptake in transportation: India has the capacity to produce 62 million metric tons (MMT) of compressed bio-gas (CBG) by fully tapping into its potential. This vast production capacity alone can sufficiently meet the entire natural gas demand of the country.³⁸ The production of bioethanol has surged approximately 20-fold, rising from 15.4 crore litres in 2012-13 to 304 crore litres in 2020-21.³⁹ Additionally, biodiesel production has nearly doubled, increasing from 10 crore litres in 2010-11 to 18.5 crore litres in 2022-23.⁴⁰ Harnessing these bioenergy sources presents a promising opportunity to replace fossil fuels in the automotive sector, thereby effectively reducing associated emissions.



Employment generation potential: The bioenergy sector is poised to become one of the largest employers in India in the renewable energy space. As per estimates, the bioenergy sector creates 12-18 direct white-collar jobs per megawatt of energy generation in comparison to 5-10 and 7-8 jobs created in the solar and wind energy sectors, respectively. This is driven by the highly labour-intensive nature of the bioenergy sector, especially due to the requirement of an additional workforce to manage biomass fuel sourcing and processing.⁴¹



Waste management: Bioenergy facilities can process organic waste materials such as agricultural residues, food scraps, animal manure, and sewage sludge. By converting these wastes into energy, the volume of waste that needs to be disposed of in landfills or incinerated is significantly reduced, helping to minimise the burden on waste management systems. Additionally, India can also consider taking advantage of landfill gas (LFG) capture to generate electricity, reduce greenhouse gas emissions, and to earn carbon emission reduction credits.⁴²

³⁶ Ministry of New and Renewable Energy, Bio Energy, 2022

³⁷ Ministry of New and Renewable Energy, Bio Energy, 2022

³⁸ Centre for Science and Environment, COMPRESSED BIOGAS (CBG) : Potential, Technology, Policy, Operations and Economics, 2023

³⁹ ORF, The Implications of India's Revised Roadmap for Biofuels: A Lifecycle Perspective, 2022

⁴⁰ Ministry of Petroleum and Natural Gas, Biodiesel, 2022

⁴¹ PwC, Fuelling India's future with bioenergy, 2023

⁴² IEA, Turning a Liability into an Asset: Landfill Methane Utilisation Potential in India, 2008

Challenges

This section highlights some of the major challenges and constraints facing bioenergy in India. Efforts to address these challenges will contribute to energy security through increased use of domestically produced ethanol and support the transition to cleaner cooking for the country's large population.



Insufficient and scattered data related to biomass availability impede project planning and cost estimation: The lack of easily accessible biomass data at the district and block level necessitates individual research, impacting project planning. As a result, key stakeholders such as government agencies at the national and state level independently analyze biomass availability, leading to uncertain accuracy, time efficiency, and cost-effectiveness of surplus biomass for bioenergy production.⁴³



In the states possessing ~75% of India's biomass availability, installed capacity is lagging potential. Only Uttar Pradesh and Maharashtra have tapped into their full biopower potential. A Comptroller and Auditor General of India (CAG) report of 2015 suggested multiple reasons for low momentum across states, including the lack of biopower policies in the majority of the states, discrepancies in the disbursement of funds by MNRE, inadequate implementation despite the availability of funds, reliance on coal rather than bagasse, and various additional factors.⁴⁴



Lack of functioning market mechanisms: The absence of well-established biomass markets with standardized pricing and trading mechanisms hampers the efficient exchange of biomass feedstock and impedes the negotiation of fair prices and long-term contracts. Combined with fragmented supply chains, this restricts trading to a few states, despite the nationwide demand for biomass and waste materials. As a result, sourcing agencies, aggregators, and processed biomass producers face challenges in connecting with potential buyers.⁴⁵



High upfront costs of bioenergy projects and lack of adequate financing mechanisms: The upfront costs of bioenergy projects are high in India. For instance, the cost of setting up a 2G biofuel ethanol plant is INR 1,000 crores, while that for a 1G plant is INR 100-200 crores. Additionally, the banks have a low understanding of assessing the viability of the bioenergy projects and hence are reluctant to finance them. Evidence suggests that even if banks finance bioenergy projects, the interest rates for loans of bioenergy projects are higher than other clean energy sources, thus impeding bioenergy production in the country.⁴⁶



High cost of biomass storage: Agricultural residues, particularly, are highly bulky and available only for some months of the year. The large storage spaces required to store enough raw materials for unaffected production throughout the year raise production costs. Biomass storage also faces issues of potential degradation during storage due to biological, and chemical reactions, resulting in losses.⁴⁷



Current Compressed Biogas pricing does not reflect the cost of different feedstocks used in production, which creates an uneven playing field for producers: The Ministry of Petroleum and Natural Gas established a pricing mechanism for Compressed Biogas (CBG) by aligning it with the prices of Compressed Natural Gas (CNG) to provide support to oil and marketing companies (OMCs). However, the Standing Committee on Petroleum and Natural Gas mentioned in its report that the current pricing structure overlooks the varying costs associated with different feedstocks used in Compressed Biogas (CBG) production. This uniform pricing approach favors certain feedstocks while discouraging others, potentially introducing bias in the market. Recognizing this concern, the Committee urged for a more

⁴³ United Nations Centre For Regional Development, Discussion paper on Potential of economic utilization of biomass waste in India - Implications towards SDGs, 2016

⁴⁴ CAG, Biomass Power, 2015

⁴⁵ United Nations Centre For Regional Development, Discussion paper on Potential of economic utilization of biomass waste in India - Implications towards SDGs, 2016; PwC India, Fuelling India's future with bioenergy, 2023

⁴⁶ PwC India, Fuelling India's future with bioenergy, 2023; International Energy Agency, India Energy Outlook 2021, 2021; United States Department of Agriculture, Biofuels Annual, 2021

⁴⁷ IEA Bioenergy, Dry matter losses during biomass storage, 2019

robust and flexible pricing strategy that considers the cost economics of CBG production using different feedstocks.⁴⁸



High tariffs for electricity from biomass power plants relative to other RE sources discourage DISCOMs from procuring biopower: The existing high tariff of electricity from biomass plants, especially relative to wind and solar power, has discouraged DISCOMs from procuring biopower. For instance, in Punjab, the tariffs on solar power have gone down to INR 2.25-2.50 per unit, while the price of biopower is INR 8, leading to low offtake.⁴⁹



Insufficient representation of female workers in the bioenergy labour force: By 2022, India's bioenergy projects had created approximately 4,30,000 direct jobs and 6,60,000 indirect jobs. However, out of these, women held only 2,50,000 (23%) jobs in the bioenergy value chain, lower than the global average of 32% for women in the renewables sector.⁵⁰

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⁴⁸ Parliament Digital Library, Review of Implementation of CBG (SATAT), 2022

⁴⁹ Mongabay, From pollution to power, what stops paddy straw from becoming the fuel of choice, 2021; MNRE – Punjab Energy Development Agency, Biomass Power Projects, 2021

⁵⁰ PwC, Fuelling India's future with bioenergy, 2023; IRENA, Renewable Energy: A gender perspective, 2019

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee bio energy.



NITI Aayog

Conducts policy analysis of energy sectors related to power, coal, petroleum and natural gas, new and renewable energy, and atomic power; and undertakes steps to improve energy planning, and data management in India



Ministry of New and Renewable Energy (MNRE)

Develops renewable and clean energy technologies, systems, products, and services in line with global standards and deploys these for energy applications for sectors like power, transport and support country's objective of achieving energy security



Bureau of Energy Efficiency

Assists in developing policies and strategies to promote energy efficiency and conservation and utilises existing resources and infrastructure in performing functions under the Energy Conservation Act



Central Electricity Authority

Acts as a technical support base for power sector stakeholders and supports policy formulation, establishing technical standards, project monitoring and human resource skilling



Central Electricity Regulatory Commission

Improves regional transmission system operations, formulates efficient tariff mechanisms, promotes power market development, and facilitates technological advancements alongside environmental and safety compliance



Ministry of Petroleum and Natural Gas

Undertakes exploration and extraction of oil and natural gas, as well as the refining, distribution, and marketing of petroleum products, alongside handling activities related to the import, export, and conservation of these products



Indian Renewable Energy Development Agency

Finances and supports self-sustainable investments in energy generation from renewable sources, and carry out promotion of energy efficiency and sustainable environmental technologies

Major policies and programs

This section highlights existing policies related to flood management and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



(Notified) National Bioenergy Programme, 2022⁵¹

The programme aims to increase bioenergy usage through:

- Supporting the installation of large biogas-, bio-CNG power- plants
- Promoting the use of pellets and briquettes power generation and non-bagasse-based power generation projects
- Setting up of family and medium size biogas installations in rural areas
- Implement Waste to Energy programme to support generation of BioCNG from urban, industrial, and agricultural waste



(Amended) National Policy on Bio-Fuels, 2022⁵²

The Policy enables the availability of biofuels in the market by:

- Proposing a target of 20% blending of ethanol in petrol Ethanol Supply Year (ESY) 2025-26
- Reducing the import of petroleum products by fostering domestic biofuel production



Sustainable Alternative Towards Affordable Transportation (SATAT)⁵³

The initiative aims to boost uptake of Compressed Bio Gas (CBG) adoption through:

- Establishing 5000 BioCNG plants with production target of 15 MMT of BioCNG by 2023-24
- Encouraging entrepreneurs to set up BioCNG plants, produce & supply BioCNG to Oil Marketing Companies (OMCs) for sale as automotive fuels



Galvanising Organic Bio-Agro Resources Dhan (GOBARdhan)⁵⁴

The scheme aims to positively impact village cleanliness and generate energy from cattle and organic waste to promote circular economy.



National Biogas and Organic Manure Programme⁵⁵

The scheme aims to provide clean cooking fuel for kitchens, lighting, and meeting other thermal and small power needs of farmers/dairy farmers /users including individual households, and to improve organic manure system based on bio-slurry from biogas plants in rural and semi-urban areas by setting up of small size biogas plants of 1 to 25 Cubic Metre capacity.

⁵¹ Ministry of New and Renewable Energy, MNRE notifies National Bio Energy Programme, 2022

⁵² Ministry of Petroleum and Natural Gas, National Policy on Biofuels-2018 Amendment, 2022

⁵³ Ministry of New and Renewable Energy, Establishment of Bio-CNG plants, 2022

⁵⁴ Ministry of Jal Shakti, GOBARdhan

⁵⁵ India Science, Technology & innovation, Programmes & Schemes, accessed on December 2023

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. These are based on reports and materials prepared by key government bodies and expert institutions like the Ministry of New and Renewable Energy and the IEA.

 Enhance investments in key public infrastructure across the bioenergy supply chain	Enhance storage and distribution assets and market mechanisms for aggregating and trading biomass - Strengthen existing biomass sourcing supply chain networks - Establish mechanisms for the efficient segregation of biomass waste - Support the development of platforms that enable biomass trading across states Advance R&D and support the development of real-time data on bioenergy sources - Harmonise and deepen data on the availability of biomass across states and districts to enable more geographically disaggregated planning - Accelerate R&D focused on more efficient fuel processing and biomass conversion technologies (e.g., by expanding existing bioenergy innovation centers to conduct research on feedstock enhancement, integration with carbon capture technologies, and waste-to-energy interventions)
 Actively promote bio-energy use-cases	Incentivise more and diverse use cases and offtakers for bio energy - Non-power uses in particular, such as through collaborations with vehicle manufacturers to support higher biofuel blending, wider uptake of bio-fertilisers in place of chemical fertilisers, and greater adoption of co-firing in thermal power plants, among others, need active promotion
 Mobilize funding for bioenergy projects	Promote innovative sources of capital to drive private investment - Credit guarantees and concessional finance can help address heightened risk perceptions from financial institutions and increase financing (e.g, through support from SIDBI's partial risk sharing facility)
 Ensure adequate feedstock supply	Promote the development of diverse and high-quality feedstocks for biofuel - Enable the use of residue oils (such as used cooking oil for biodiesel) as well as algal feedstock and seaweed (for ethanol production) to diversify raw material availability for biofuel production

Recommended readings

MNRE and ASCI. (2021). *Final Report on Evaluation Study for the Assessment of Biomass Power and Bagasse Cogeneration Potential in the Country.*

<https://cdnbbsr.s3waas.gov.in/s3716e1b8c6cd17b771da77391355749f3/uploads/2023/05/2023053179.pdf>

IEA. (2021). *Implementation of Bio energy in India*

https://www.ieabioenergy.com/wp-content/uploads/2021/11/CountryReport2021_India_final.pdf

IEA. (2021). *India Energy Outlook.*

https://iea.blob.core.windows.net/assets/1de6d91e-e23f-4e02-b1fb-51fdd6283b22/India_Energy_Outlook_2021.pdf



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